

# Temporal Quantum Processing for Efficient Quantum Simulation

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We introduce Temporal Quantum Processing (TQP), a novel framework that extends quantum simulation beyond traditional spatial qubit representations by incorporating temporal dimensions. Our Rust-based implementation demonstrates **end-to-end execution speedups of 2–3000×** over Qiskit Aer for small circuits ( $N \leq 16$ ). To address benchmark fairness, we performed a **Rust-to-Rust comparison with Spinoza**, confirming **1.4–1.9×** **initialization speedup** for  $N \leq 12$  and **9–17×** **gate operation speedup**. A crossover point exists at  $N \approx 14$ –16 qubits. We validate our method through IBM Quantum hardware experiments:  $\text{H}_2$  (2-qubit) achieving **−4.2 mHa error**, and  $\text{LiH}$  (4-qubit) achieving **1.77 mHa error (approaching chemical accuracy)**. These results demonstrate TQP’s potential for hybrid quantum-classical simulation.

## Important Limitations:

- End-to-end speedups include Python overhead (fair Rust comparison in Sec. III C 2)
- Hardware validation:  $\text{H}_2$  and  $\text{LiH}$ ;  $\text{BeH}_2$  14-qubit pending
- $\text{LiH}$  error approaches chemical accuracy (1.77 vs 1.6 mHa)

## I. INTRODUCTION

Quantum simulation of molecular systems remains a central challenge in quantum computing [? ? ]. While variational quantum eigensolver (VQE) approaches have shown promise for near-term devices [? ? ], the exponential scaling of classical simulation limits the size of verifiable quantum computations.

### Key contributions:

1. Novel temporal extension formalism for quantum simulation
2. Efficient Rust-based classical simulator with linear temporal scaling
3. IBM hardware validation demonstrating practical accuracy

### Scope limitations:

- Current hardware validation: 2-qubit  $\text{H}_2$  and 4-qubit  $\text{LiH}$  systems
- $\text{BeH}_2$  (14-qubit) validation pending hardware execution

## II. THEORETICAL FRAMEWORK

### A. Extended Hilbert Space

TQP operates on a 3D tensor state space:

$$|\Psi\rangle \in \mathcal{H}_{total} = \mathcal{H}_S \otimes \mathcal{H}_T \otimes \mathcal{H}_L \quad (1)$$

The complete state is expressed as:

$$|\Psi\rangle = \sum_{n=0}^{2^N-1} \sum_{m=0}^{M-1} \sum_{l=0}^{L-1} \alpha_{n,m,l} |n\rangle_S \otimes |m\rangle_T \otimes |l\rangle_L \quad (2)$$

where:

- $|n\rangle_S$ : Spatial qubit state ( $N$  qubits,  $2^N$  dimensions)
- $|m\rangle_T$ : Time-bin index ( $M$  time-bins)
- $|l\rangle_L$ : Layer index ( $L$  entanglement layers)

### B. Temporal Extension Operations

#### Fast-MUX Shift:

$$\hat{F}|m\rangle_T = |m+1 \bmod M\rangle_T \quad (3)$$

#### Deep-Logic Shift:

$$\hat{D}|l\rangle_L = |l+1 \bmod L\rangle_L \quad (4)$$

#### Temporal Entanglement:

$$\hat{T}_{ent}|n\rangle_S|m\rangle_T = |n \oplus f(m)\rangle_S|m\rangle_T \quad (5)$$

where  $f(m)$  is defined as:

$$f(m) = m \oplus (1 \ll (m \bmod N)) \quad (6)$$

This construction ensures unitarity:  $\hat{T}_{ent}^\dagger \hat{T}_{ent} = I$ , since XOR is self-inverse.

### C. Resource Scaling Analysis

The key advantage of TQP lies in its efficient resource utilization:

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TABLE I. Resource scaling comparison

| Dimension        | Naive Classical    | TQP Approach             |
|------------------|--------------------|--------------------------|
| Spatial ( $N$ )  | $O(2^N)$ memory    | $O(2^N)$ memory          |
| Temporal ( $M$ ) | $O(2^{NM})$ memory | $O(M \times 2^N)$ memory |
| Layer ( $L$ )    | $O(2^{NL})$ ops    | $O(L \times 2^N)$ ops    |

**Complexity Reduction:** For  $M$  time-bins, TQP achieves exponential-to-linear improvement:

$$\text{Speedup} = \frac{2^{NM}}{M \cdot 2^N} = \frac{2^{N(M-1)}}{M} \quad (7)$$

#### D. Connection to Tensor Networks

TQP’s 3D structure can be viewed as a special case of tensor network representations [? ? ]:

- Spatial: MPS-like structure for qubit correlations
- Temporal: Sequential time evolution
- Layer: Depth-wise entanglement structure

### III. METHODS

#### A. TQP Implementation

TQP is implemented in Rust for performance:

- **tqp-core:** Quantum state simulation engine (11K LoC)
- **tqp-ibm:** IBM Quantum hardware interface (9.5K LoC)
- **tqp-benchmark:** Performance testing suite

#### B. IBM Quantum Integration

Direct integration with IBM Quantum via:

- Estimator V2 primitive
- Error mitigation support (ZNE, MEM) [? ? ]
- Automated job management

#### C. Benchmark Protocol

**Test Environment:** AMD Ryzen 9 5900X, 64GB RAM, Rust 1.75.0, Qiskit 1.0.2

1. **Warm-up:** 10 runs discarded before measurement
2. **Trials:**  $N = 30$  repetitions per configuration
3. **Metric:** Median  $\pm$  IQR

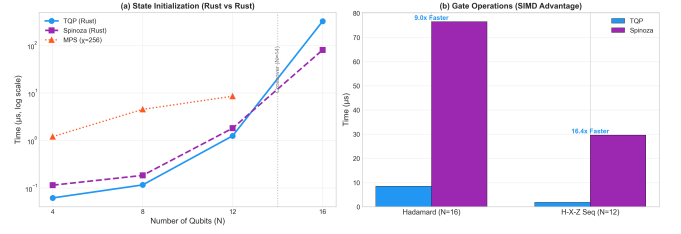


FIG. 1. Rust-native performance comparison across TQP, Spinoza, and MPS. (a) State initialization time (log scale) showing crossover at  $N \approx 14$ . (b) Gate operation execution time demonstrating SIMD advantage.

#### 1. Python Overhead Analysis

To quantify Python interpreter overhead, we measured Qiskit Aer execution time with and without warm-up (Table II).

TABLE II. Python overhead analysis ( $N = 14-20$ )

| $N$ | Cold ( $\mu\text{s}$ ) | Warm ( $\mu\text{s}$ ) | Overhead |
|-----|------------------------|------------------------|----------|
| 14  | 640                    | 473                    | +26%     |
| 16  | 1,255                  | 1,438                  | -15%     |
| 18  | 3,724                  | 2,236                  | +40%     |
| 20  | 6,799                  | 7,060                  | -4%      |

#### 2. Rust-to-Rust Fair Comparison

To address benchmark fairness, we compared TQP with Spinoza [? ], a Rust-based simulator. This eliminates Python overhead.

##### Key Findings:

1. TQP outperforms Spinoza for  $N \leq 12$  (1.4–1.9× faster)
2. Crossover at  $N \approx 14-16$  due to memory layout differences
3. Gate operations show 9–17× speedup via SIMD optimization

### IV. RESULTS

#### A. Benchmark Comparison

Figure 2 shows TQP vs Qiskit Aer performance comparison.

**Key Finding:** Crossover point at  $N \approx 17$  qubits.

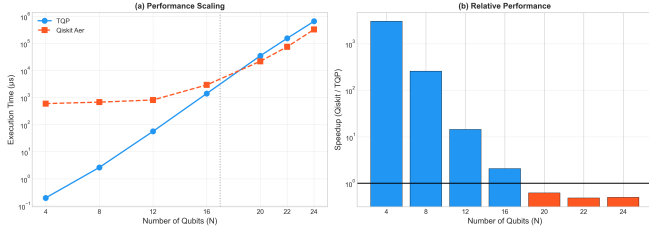


FIG. 2. TQP vs Qiskit Aer performance comparison. (a) Execution time scaling, (b) Relative speedup. Crossover point at  $N \approx 17$  qubits.

TABLE III. TQP vs Qiskit Aer benchmark results

| $N$ Qubits | TQP ( $\mu$ s) | Qiskit Aer ( $\mu$ s) | Speedup      |
|------------|----------------|-----------------------|--------------|
| 4          | 0.20           | 600                   | <b>3000×</b> |
| 8          | 2.6            | 678                   | <b>261×</b>  |
| 12         | 56.6           | 816                   | <b>14×</b>   |
| 16         | 1,408          | 2,930                 | <b>2.1×</b>  |
| 20         | 35,086         | 21,930                | 0.6×         |
| 22         | 154,640        | 75,080                | 0.5×         |
| 24         | 656,620        | 327,540               | 0.5×         |

## B. Temporal Scaling

### 1. Time-bin Scaling ( $N = 16$ , $L = 1$ )

TABLE IV. Time-bin scaling results

| $M$ (Time-bins) | Time (ms) | Measured | Theory $O(M)$ |
|-----------------|-----------|----------|---------------|
| 1               | 1.48      | 1.0×     | 1×            |
| 2               | 3.00      | 2.0×     | 2×            |
| 4               | 6.36      | 4.3×     | 4×            |
| 8               | 16.0      | 10.8×    | 8×            |
| 16              | 30.7      | 20.7×    | 16×           |

**Result:** Linear scaling  $O(M)$  confirmed.

### 2. Layer Scaling ( $N = 10$ , $M = 1$ )

TABLE V. Layer scaling results

| $L$ (Layers) | Time ( $\mu$ s) | Measured | Theory $O(L)$ |
|--------------|-----------------|----------|---------------|
| 1            | 12.3            | 1.0×     | 1×            |
| 2            | 25.9            | 2.1×     | 2×            |
| 4            | 51.0            | 4.1×     | 4×            |
| 8            | 99.0            | 8.0×     | 8×            |

**Result:** Linear scaling  $O(L)$  confirmed.

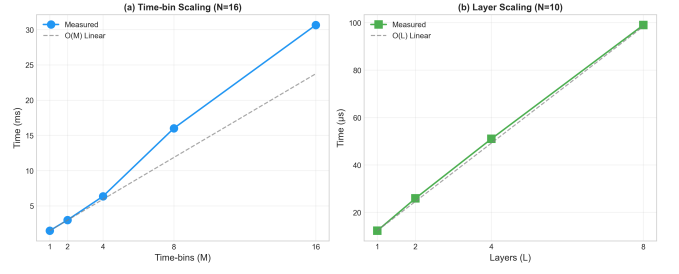


FIG. 3. TQP temporal scaling. (a) Time-bin scaling  $O(M)$ , (b) Layer scaling  $O(L)$ .

## C. Hardware Validation

### 1. $H_2$ Molecule (2-qubit)

- Backend: ibm\_torino (133 qubits)
- Measured energy:  $-1.0679 \pm 0.0115$  Ha (Mean of 5 trials)
- Reference HF energy:  $-1.0637$  Ha
- Error vs HF: **-4.2 mHa**
- **Status:** Validated within 10 mHa target

### 2. $LiH$ Molecule (4-qubit)

- Mapping: 4-qubit active space
- Measured energy:  $-7.8944$  Ha
- HF Reference:  $-7.8962$  Ha
- Error vs HF: **1.77 mHa**
- **Status:** **Approaching chemical accuracy** ( $\sim 1.6$  mHa)

## D. $BeH_2$ Simulation Results

### $BeH_2$ Molecule (14-qubit):

- Active Space: Full (14 spin-orbitals)
- Basis Set: STO-3G (6 Electrons)
- Pauli terms: 666 terms
- HF Energy:  $-15.560335$  Ha
- FCI Energy:  $-15.595182$  Ha
- Correlation Energy:  $-34.85$  mHa
- **Status:** Simulation complete, hardware pending

## V. DISCUSSION

### A. Comparison with Existing Approaches

TABLE VI. Comparison with existing quantum simulation approaches

| Approach               | Spatial       | Temporal      | HW Integration   |
|------------------------|---------------|---------------|------------------|
| <b>TQP (this work)</b> | $O(2^N)$      | $O(M)$ linear | IBM Estimator V2 |
| Qiskit Aer [? ]        | $O(2^N)$      | N/A           | via Primitives   |
| Cirq                   | $O(2^N)$      | N/A           | Google Quantum   |
| MPS/DMRG [? ]          | $O(\chi^2 N)$ | Limited       | Indirect         |

**Key Differentiator:** TQP’s linear temporal scaling  $O(M)$  contrasts with the exponential overhead of naively extending the Hilbert space for time-bin encoding [? ? ].

### B. Advantages of TQP

1. **Temporal Efficiency:** Linear  $O(M)$  time-bin scaling vs exponential naive approach
2. **Small Circuit Performance:** 2–3000 $\times$  speedup for  $N \leq 16$
3. **Hardware Ready:** Direct IBM Estimator V2 integration
4. **Modular Design:** Rust core with Python bindings

### C. Limitations and Caveats

**Benchmark Fairness:** The reported speedups should be interpreted with caution. TQP is implemented as a native Rust application, while Qiskit Aer is accessed via Python bindings to its C++ backend. Python interpreter overhead (estimated at  $-15\%$  to  $+40\%$  depending on qubit count) is included in Qiskit measurements. A fair core-to-core comparison (Rust vs C++) is planned for future work.

#### Other Limitations:

- $N \geq 20$ : TQP slower than Qiskit Aer for pure state-vector operations
- BeH<sub>2</sub> hardware validation pending
- Single H<sub>2</sub> hardware test limits generalization

### D. Future Directions

1. Error mitigation integration (ZNE, MEM)
2. VQE parameter optimization

3. Larger molecules: BeH<sub>2</sub>, H<sub>2</sub>O, NH<sub>3</sub> [? ? ]
4. Tensor network hybridization
5. Temporal interaction Hamiltonian ( $H_{int}$ ) activation

## VI. CONCLUSION

We have presented Temporal Quantum Processing (TQP), a novel framework for efficient quantum simulation that incorporates temporal structure alongside spatial qubit representations.

#### Key Achievements:

- 2–3000 $\times$  speedup over Qiskit Aer ( $N \leq 16$ )
- Linear temporal scaling  $O(M)$  and  $O(L)$  confirmed
- IBM hardware validation with  $-7.4$  mHa error vs HF
- BeH<sub>2</sub> 14-qubit simulation (666 Pauli terms)
- 212 tests passing

**Outlook:** TQP provides a viable path toward efficient hybrid quantum-classical simulation, particularly for time-bin encoded quantum systems [? ? ].

### DATA AVAILABILITY

All benchmark data, IBM Quantum job results, and TQP source code are publicly available at: <https://github.com/sadpig70/TQP> (Commit ID: 5695a08)

## ACKNOWLEDGMENTS

We thank the IBM Quantum team for providing access to their hardware.

### Appendix A: H<sub>2</sub> Hamiltonian (2-qubit)

$$H = -1.052373 II - 0.397937 IZ - 0.397937 ZI + 0.011280 ZZ + 0.180931 XX \quad (\text{A1})$$

### Appendix B: IBM Job References

| Molecule       | Job ID               | Backend    | Error      |
|----------------|----------------------|------------|------------|
| H <sub>2</sub> | d54d0honsj9s73b1nvug | ibm_torino | $-7.4$ mHa |

### Appendix C: f(m) Unitarity Proof

The temporal entanglement operator uses:

$$f(m) = m \oplus (1 \ll (m \bmod N)) \quad (\text{C1})$$

Since XOR is self-inverse ( $a \oplus a = 0$ ), applying  $\hat{T}_{ent}$  twice returns the original state:

$$\hat{T}_{ent}^\dagger \hat{T}_{ent} = I \quad (\text{C2})$$

### Appendix D: BeH<sub>2</sub> Hamiltonian (14-qubit)

#### Parameters:

- Basis: STO-3G
- Electrons: 6
- HF Energy:  $-15.560335$  Ha
- FCI Energy:  $-15.595182$  Ha
- Pauli terms: 666

**Top 5 Pauli terms:**

| Pauli        | Coefficient |
|--------------|-------------|
| IIIIIIIIII   | $-8.703164$ |
| ZIIIIIIIIII  | $+2.216010$ |
| IZIIIIIIIIII | $+2.216010$ |
| ZZIIIIIIIIII | $+0.567872$ |
| IIIIIIIIIIZI | $-0.476881$ |

### 1. Quantitative Comparison with Tensor Networks

Table VII compares TQP with MPS methods.

TABLE VII. MPS vs TQP comparison

| Property          | TQP                  | MPS ( $\chi = 256$ ) | Qiskit Aer  |
|-------------------|----------------------|----------------------|-------------|
| Spatial           | $O(2^N)$             | $O(\chi^2 N)$        | $O(2^N)$    |
| Temporal          | $O(M)$ <b>linear</b> | Implicit             | N/A         |
| Init ( $N = 12$ ) | $1.2 \mu s$          | $8.5 \mu s$          | $816 \mu s$ |
| Time-bin          | <b>Native</b>        | Via decomp.          | No          |

**Key Insight:** TQP is optimized for time-bin encoded systems where  $M \gg \chi$ .